Humanoid Robot Research —Thesis Briefing

Vision-Language-Action policies on the Unitree G1 EDU: context, proposed thesis, and open questions

30 October 2026

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Humanoid robot research —Master's thesis workspace

Research workspace for a Master's thesis on **Vision-Language-Action (VLA) policies on a Unitree G1 humanoid**, in a robotics lab at Nankai University. Lab source material is in Chinese; everything produced here is in English.

What is happening

The project started from a Chinese sales brochure and an assumption that the work would be control and perception engineering on a Unitree **R1**. Both turned out to be wrong:

- The lab's robot is a **G1 EDU**, not an R1. The brochure describes a *workbench* from a different product line and names G1 zero times.
- Day-to-day supervision covers the **VLA / decision-making layer** only. Control algorithms and simulation cannot be supported in-group.

So the direction moved from "learn the motion stack"to a question that is both supervisable and runnable on the hardware that is actually confirmed present:

> **A VLA policy that is accurate offline is not necessarily usable on a robot, because the robot keeps moving while the policy thinks. Where does the latency budget go, and what does spending it differently cost in task success?**

Full reasoning, including the two directions that were considered and deferred, is in `01-thesis-proposal.md`.

What needs to be done

Blocked on the lab —five questions, best sent as one numbered message. The most important is *which VLA stack the group actually uses*, because the thesis should target their stack.

Not blocked —start now. Three pieces of work need no hardware, no permissions and no answers: build the latency timestamping harness, write the task-suite specification, and draft the related-work section. Together they de-risk everything that follows.

Both lists, with the reasoning and the order to do them in, are in `02-open-questions.md`.

Documents

File	What it is
`00-research-context.md`	Established facts —hardware, supervision, constraints. Every claim labelled `[CONFIRMED]` / `[VENDOR]` / `[UNVERIFIED]`
`01-thesis-proposal.md`	The proposed thesis: question, why this and not the alternatives, method, risks, PhD arc
`02-open-questions.md`	Decision gates and the work that can start immediately
`reference/bench-brochure-translation.md`	Full English rendering of the bench brochure. Background only

File	What it is
reference/sport-services-api.md	Motion API reference (`sport` service, `LocoClient`), read from SDK source
docs/	Generated PDF and DOCX of the proposal, for sharing with the lab

Generating the shareable documents

The PDF and DOCX in `docs/` are built from the Markdown —edit the Markdown, never the generated files:

```
./build-docs.sh
```

Requires `pandoc`; the PDF step additionally requires a LaTeX engine (`xelatex`).

Conventions

- **Claims are labelled.** Nothing moves from `[UNVERIFIED]` to `[CONFIRMED]` without a source or a direct statement from the lab. Literature numbers are not cited until checked against the papers.
- **Hardware is never assumed.** No plan depends on hardware not confirmed present. This is why the task suite is tiered.
- **Safety.** The humanoid is never operated alone —two people present, e-stop within reach.

A note on sources

The vendor brochure underlying the translation is Unitree Robotics material and is **not redistributed here**; it is kept locally and excluded from version control. Personal notes, supervision notes and lab machine configuration are likewise local-only.

Research context —established facts

Everything in this file is treated as settled. It is the shared baseline for 01-thesis-proposal.md and 02-open-questions.md. Each claim is labelled so that nothing unverified is silently promoted to fact:

- **[CONFIRMED]** —stated directly by the lab, or read off the hardware.
- **[VENDOR]** —from Unitree product documentation or the bench brochure.
- **[UNVERIFIED]** —working belief from prior reading. **Must be checked before citing.**

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1. The correction that reframes the project

The project began from a 6-page Chinese sales brochure, translated in full in reference/bench-brochure-translation.m

Two things about that brochure matter, and both are easy to get wrong:

1. **It describes a workbench, not a robot.** The product is model SXT-LK-R1-1: a 1450 × 800 × 1300 mm bench with a protective gantry, AC power and e-stop panel, a cut-away motor display, a host PC, and a microphone/speaker array. The humanoid itself never appears in any equipment list in the document. **[VENDOR]**
2. **It describes the wrong robot line.** The brochure names **R1** five times and **G1** zero times. **[VENDOR]**

The lab's actual robot is a **G1**. **[CONFIRMED]** So the brochure is background material about a product line the thesis does not use. It is kept in `reference/` for traceability and is not the basis for anything.

Why this matters: an entire earlier plan —motor fundamentals → motion control → LiDAR/SLAM → gripper/ACT capstone —was derived from this brochure. It was built on the wrong robot *and* at the wrong layer (see §3). That plan is superseded.

2. Hardware actually in the lab [CONFIRMED]

Item	Status	Note
Unitree **G1** humanoid	Present, **EDU**	The machine currently in use
—its hands	**Fake / decorative**	Non-functional end effectors. Drives the task design in §5 of the proposal
A **second G1** with 17-joint hands	Present	Availability to this project **not yet confirmed**
Three-finger movable hand	Present, **detached**	Permitted in principle to attach to the fake-hand G1; not yet done
Robot dogs (quadrupeds)	Present	All EDU **except** the smallest, which is not EDU

EDU is the decisive label. Only EDU editions support secondary development —own code, own algorithms, SDK and ROS 2 access. Basic and Air editions ship locked and are demonstration-only. **[VENDOR]** The lab being almost entirely EDU is what makes research possible here at all.

G1 EDU configuration [VENDOR]: NVIDIA Jetson **Orin NX** high-compute module as standard, 3D LiDAR and depth camera included, optional Dex3-1 three-finger dexterous hand (7 DoF per hand), knee torque to 120 N · m, arm payload 3 kg. Non-EDU G1 drops the Orin module and rates lower (90 N · m, 2 kg).

Network [CONFIRMED]

The host machine sits on Unitree's standard `192.168.123.x` robot subnet with a manual/static address, `255.255.255.0` mask and a `.1` gateway —a conventional bring-up. The number that matters for this thesis is the link speed: **100 / 100 Mbps**.

Why it matters: 100 Mbps is comfortable for control messages and marginal for streaming camera frames off-robot. It is a real argument for keeping policy inference **onboard**, and it makes "onboard vs off-board" a measurable question rather than a preference. This is used directly in the proposal.

Safety rules [CONFIRMED]

- Robot dog: may be worked on alone.
- **Humanoid: two people must be present.** E-stop within reach.
- Extra caution on the dexterous-hand G1, which is rarely used by anyone in the lab.

Why it matters beyond safety: the two-person rule makes robot time a *scheduled, scarce resource*. Any plan that needs long unsupervised sessions is not executable here, which is a further argument for short, scripted, instrumented runs.

3. Supervision —what can and cannot be guided [CONFIRMED]

Day-to-day guidance is available for the **decision-making layer**: large-model agents integrated with robots, i.e. **Vision-Language-Action (VLA)**.

Guidance is **not** available for:

- low-level control algorithms,
- simulation engineering (MuJoCo / Isaac Gym),
- Action Grid, which is outside the group's work.

Why this is the single most important constraint in the project. A thesis is defensible largely because a supervisor can judge it. Work that sits outside the supervisor's area has to be carried alone, and a non-robotics student carrying a control or simulation thesis alone is the main foreseeable way this fails. The thesis therefore lives in the VLA / decision / evaluation layer, and treats any simulation or control work as optional and self-supported.

4. Technical reference carried forward

The motion API is documented in `reference/sport-services-api.md`: DDS service `sport`, client class `LocoClient`, getters in 7001 – 7006, setters in 7101 – 7107, control switching at 7110/7111, plus friendly wrappers (`Damp`, `StandUp`, `Move`, `WaveHand`, ⋯). "Deploying a predefined motion"means selecting an FSM or task id and sending it.

Why it is kept but demoted: the thesis needs a way to make the robot move safely and repeatably, and this is it. But the API surface is genuinely small, so *using* it is a means, not a contribution. Treating it as the thesis was the earlier mistake.

R1 and G1 share `unitree_sdk2` / `unitree_sdk2_python`, CycloneDDS and the `192.168.123.x` layout, but differ in joint count, URDF and partly in API —so R1 examples are a guide, not a drop-in. **[VENDOR]**

5. Literature figures —[UNVERIFIED], do not cite yet

Carried from earlier reading, recorded so they are not lost, flagged so they are not misused:

- ~50 Hz onboard closed-loop VLA demonstrated on G1 with Orin NX.
- **POT-VLA** formalised "object-state divergence"and reported 39/80 → 71/80 success.
- **DEED** reported data-efficient post-training taking a task 0% → 32%, framing it as a systems-integration problem rather than an architecture problem.

None of these has been checked against the source papers. They are not load-bearing: the proposal in `01-thesis-proposal.md` is written so that its motivation stands without them. Verifying them is task V1 in `02-open-questions.md`.

6. Standing rules

- Never run the humanoid alone; two people, e-stop in reach.
- Never propose work that depends on hardware not confirmed present.
- Never assume in-group support for control or simulation.
- The thesis must answer a question with a measurable result. A learning log is not a thesis.
- All repository documents in English.
- Verify literature numbers before citing.

Thesis proposal

Status: recommended direction, ready to take to the supervisor. Not yet agreed with the lab. Baseline facts in `00-research-context.md`; everything still to be confirmed is in `02-open-questions.md`.

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1. The decision, and why

Working title

> **Latency-Bounded Deployment of Vision-Language-Action Policies on a Humanoid Robot: action representation, compute placement, and closed-loop task success on the Unitree G1 EDU**

Short form: *How fast must a humanoid think?*

The one-sentence question

A VLA policy that is accurate offline is not necessarily usable on a robot, because the robot keeps moving while the policy thinks —so where is the latency budget, and what does spending it differently cost in task success?

Why this question and not another

Three candidate directions were on the table. All three sit inside the supervisor's VLA / decision-making layer, so all three are supervisable. They differ almost entirely in **how many unconfirmed things they depend on**, and that is what decides it.

	A. Latency & action representation *(recommended)*	B. Object-state divergence	C. Data curation for post-training
Inside supervisor's area	Yes	Yes	Yes
Needs working hands	**No** (see §3)	Partly —push tasks only	**Yes**
Needs demonstration data	**No**	No	**Yes**, must be collected
Needs post-training compute	**No**	No	**Yes**, unconfirmed
Needs RGB-D perception pipeline	No	**Yes**	Partly
Needs simulation	No	No	No
Fails outright if a dependency is missing	**No** —see §6	Possibly	**Yes**
Plays to a non-robotics student's strengths	**Strongly**	Partly	Partly

The deciding argument. Every one of the three options needs the same thing first: *a VLA policy actually running on this robot.* Nobody in the lab has confirmed that exists yet. Options B and C treat that bring-up as unpaid setup work before the real thesis begins. Option A **makes the bring-up itself the object of study**

—the measurement *is* the contribution. That is the difference between a thesis that starts producing results in month two and one that is still setting up in month six.

The second argument is about who is writing it. This is not a control-engineering thesis and should not pretend to be. Its contribution is an evaluation protocol, a measurement harness, a failure taxonomy and a deployment guideline —methodology and systematic characterisation. That is squarely what this student is good at, and it is a recognised class of contribution in robotics rather than a consolation prize.

Why B and C are not discarded. They are the natural follow-ons. Option A produces the instrumented platform and the failure taxonomy that B needs, and the latency/representation baseline that C needs. Sequenced as A → B → C, this is a Master's thesis with a PhD programme behind it rather than three competing ideas. See §8.

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2. Research questions

- **RQ1 —Where does the time actually go?** How does end-to-end closed-loop latency decompose across image capture, preprocessing, policy inference, action decoding, and command transmission on the G1 EDU, and which stage dominates?
- **RQ2 —What does the action representation cost?** How do discrete action tokens versus a continuous action head, and different action-chunk lengths, trade inference latency against closed-loop task success?
- **RQ3 —Is there a latency cliff?** As control latency is increased in controlled steps, does task success degrade smoothly or collapse past a threshold —and does that threshold move with task speed and with compute placement (onboard Orin NX vs off-board workstation over the lab's 100 Mbps link)?

RQ1 is descriptive and essentially cannot fail. RQ2 is the main experiment. RQ3 is the one most likely to produce a result worth publishing on its own, because a measured latency threshold is exactly the kind of number practitioners need and rarely have.

3. Task suite —designed around the hands problem

The G1 in daily use has **non-functional hands**. Rather than make the thesis wait on an unconfirmed hand, the task suite is **tiered**, and the thesis is complete at Tier 1.

Tier	Tasks	Needs	Status
1 —non-prehensile	reach-to-target, point-at-named-object, push-object-to-goal, nudge-without-toppling	Arms and vision only	**Runnable on confirmed hardware**
2 —simple grasp	pick-and-place of a graspable object	Three-finger hand attached, or access to the 17-joint G1	Gated on Q1/Q2
3 —dexterous	in-hand reorientation, multi-object sequencing	17-joint hand, sustained access	Unlikely; out of scope

Why tiering rather than picking one. It removes the project's single largest dependency without giving up ambition. Tier 1 answers all three research questions on its own —latency and action representation are not

properties of the gripper. If a hand becomes available, Tier 2 adds a generalisation result; if it never does, nothing is lost. This is the main reason this proposal is low-risk, and it is why the recommendation is Option A *as refined here* rather than Option A as originally framed, which assumed manipulation.

Why non-prehensile tasks are scientifically adequate. Pushing is a genuine closed-loop control problem with contact dynamics and continuous visual feedback —the object keeps moving after contact, so stale actions have visible consequences. That is precisely the regime where latency should matter, which makes Tier 1 arguably a *better* probe for RQ3 than slow pick-and-place.

4. Method

1. **Instrument before changing anything.** Build a timestamping harness that tags each stage of the loop and logs to a single clock. Nothing else in the thesis is trustworthy without this, and it is buildable before any hardware access.
2. **Establish a safe, repeatable motion substrate** using the documented `sport` service and `LocoClient` (see `reference/sport-services-api.md`), with e-stop procedure and a fixed reset routine between trials.
3. **Bring up one VLA policy** on the model stack the group already uses —this is gate G1 in `02-open-questions.md`, and the choice of model follows the group, not the other way round.
4. **Measure RQ1** —latency decomposition over repeated trials, per task, per tier.
5. **Measure RQ2** —hold task, scene and policy weights fixed; vary action representation and chunk length; record latency and success.
6. **Measure RQ3** —inject controlled additional latency in steps and locate the degradation curve; repeat with compute placed onboard and off-board.
7. **Build the failure taxonomy** from logged trials —including whether a failure was a *policy* error or a *timing* error. This distinction is the thesis's conceptual contribution and the bridge to Option B.

Metrics: per-stage latency distributions (not just means —tails are what break control), end-to-end loop rate, task success rate, and time-to-completion on successful trials. Every reported number tied to a logged trial id.

5. Contributions claimed

1. A reproducible **evaluation protocol and open harness** for latency-resolved VLA evaluation on a commercial humanoid.
2. An **empirical characterisation** of the latency/success frontier on real hardware.
3. A **deployment guideline**: which action representation and chunk length under which compute budget and placement.
4. A **failure taxonomy** separating policy error from timing error.

These are deliberately modest and checkable. A Master's thesis is judged on whether the claim is defended, not on how large the claim is.

6. Why this cannot fail outright

The largest technical risk is that onboard inference on the Orin NX proves impractical —too slow, or the module is not accessible.

That outcome is itself RQ3's result. If inference will not run usefully onboard, the thesis reports the measured reason and moves the policy off-board over the 100 Mbps link, where the added network latency

becomes a second experimental condition rather than a dead end. The comparison was always part of the design, so the failure mode is already a chapter.

This is the main structural reason to prefer this direction. Options B and C have failure modes that produce no result.

7. Risks, honestly

Risk	Severity	Mitigation
No VLA stack exists in the group yet	**High**	Gate G1. If none, adopt a published open model with documented Jetson deployment; budget bring-up as thesis work, not overhead
Orin NX not accessible / not installed	High	Gate G2. Off-board condition (§6)
Robot time scarce (two-person rule)	Medium	Short scripted runs; harness developed and dry-run off-robot; batch trials per session
Non-robotics background slows bring-up	Medium	Thesis contribution is methodology, not control; keep the motion substrate to documented high-level API only
Hands never become available	Low	Tier 1 is sufficient (§3)
Evaluation framed as "merely engineering"	Medium	Lead with the RQ3 threshold result, which is a scientific claim, not an engineering report
Unverified literature numbers propagate	Low	Task V1; proposal does not rest on them

8. Why this is a good base for a PhD

- **The problem is live.** Real-time, reliable embodied decision-making is a central open problem in deployed VLA, not a niche. Efficiency and latency are where the field is actively moving.
- **It transfers.** A measurement-and-evaluation skill set for VLA systems is portable to any group working on embodied foundation models —it does not lock the student to this lab, this robot, or this model.
- **It has a natural three-chapter arc.** A (this thesis: measure the frontier) → B (act on it: detect object-state divergence and recover within the latency budget) → C (reduce the cost: data-efficient post-training against the measured baseline).
- **It produces artefacts.** An open harness, a protocol and a task suite are citable and reusable, which is disproportionately valuable in a PhD application.
- **Honest counterweight:** benchmark and evaluation papers are sometimes read as lower-novelty than new-architecture papers. The defence is RQ3 —a measured latency threshold with a causal story is a finding, not a report. Keep that result in front.

9. Indicative sequence

Deliberately expressed as phases, not dates, because the gates in `02-open-questions.md` control the pace and several are outside the student's control.

Phase	Work	Depends on
0	Agree direction with supervisor; verify literature (V1); draft related-work section	Nothing —**startable now**
1	Build timestamping harness and task-suite spec; dry-run off-robot	Nothing —**startable now**
2	Robot access, safety induction, motion substrate, first teleoperated/ scripted runs	G3, G4
3	VLA bring-up on the group's stack	G1, G2
4	RQ1 and RQ2 experiments	Phase 3
5	RQ3 latency sweep, onboard vs off-board	Phase 4
6	Failure taxonomy, writing, (optional) Tier 2 extension	Phase 5

Phases 0 and 1 need no hardware, no answers and no permissions. That is where to start today; it is the fastest available path and it de-risks everything after it.

Open questions and decision gates

What is still unknown, what each answer changes, and what can proceed regardless. Ordered by how much they block, not by how interesting they are.

A **gate** blocks work until answered. A **task** can be done by the student alone, now.

Gates —ask the lab

G1. Which VLA stack does the group actually use?

GR00T, OpenVLA, $\pi 0$, or something in-house?

Why it is first. The thesis targets the group's stack, not a stack chosen in isolation —that is what makes it supervisable and what makes the result useful to the lab. It also decides whether action representation is even a free variable: some stacks fix it, which would reshape RQ2.

If there is no stack yet: that is not a blocker but it is a scope change. Bring-up of a published open model with documented Jetson deployment then becomes legitimate thesis work rather than unpaid overhead, and the timeline in §9 of the proposal shifts by roughly a phase.

G2. Is the Orin NX physically installed and accessible?

Not "standard on EDU in principle"—actually present, with an account, SSH, and a working ROS 2 / SDK environment.

Why it matters. It decides whether the headline condition is onboard or off-board inference. **It does not block the thesis** —§6 of the proposal turns a negative answer into an experimental condition —but it changes which result leads.

G3. Which G1 is available to this project, and is a hand available?

Is the 17-joint-hand unit the same machine or a separate one, and can this student use it? Is the three-finger hand cleared for attachment *in practice*, and who supervises that?

Why it matters. It selects the task tier (proposal §3). **Tier 1 proceeds regardless**, so this gate controls ambition, not viability. Worth asking early anyway, because attaching a rarely-used dexterous hand will need scheduling and a second person.

G4. What does robot access look like in practice?

Account on the bench PC, robot network details, and —given the two-person rule —how sessions are booked and who the second person is.

Why it matters. The two-person rule makes robot time the scarcest resource in the project. The experimental design (short, scripted, batched runs) depends on knowing the real shape of it.

G5. What are the formal thesis requirements?

Novelty bar, timeline, credit structure, and whether a publication is expected.

Why it matters. Whether a publication is expected changes how hard to push RQ3 and whether to target a venue. Asking it early is normal and avoids a late surprise.

G6. What compute is available for training, if any?

Workstation GPU, cluster access, or none.

Why it matters. Only lightly for this thesis, which is inference-side. It matters a lot for the Option C follow-on, so it is worth knowing now rather than later.

Tasks —no permission needed, start now

V1. Verify the three literature figures

The ~50 Hz onboard figure, POT-VLA's 39/80 → 71/80, and DEED's 0% → 32% are all recorded as **[UNVERIFIED]** in `00-research-context.md`. Find the papers, confirm or correct the numbers and the exact hardware each used.

Why now. They shape how the related-work section is argued, and an uncorrected number in a thesis draft is far more expensive to remove later. The proposal is deliberately written not to depend on them, so this is cleanup, not a blocker.

T1. Build the timestamping harness

Stage-tagged logging to a single clock, dry-run against recorded data or a stub policy.

Why now. It needs no robot, and nothing measured later is trustworthy without it. This is the single highest-value thing available today.

T2. Write the task-suite specification

Tier 1 tasks from proposal §3: initial conditions, success criteria, reset procedure, trial count.

Why now. Success criteria written *before* seeing results is what separates an experiment from a demo, and it is the part a supervisor can review without any hardware.

T3. Draft the related-work section

Latency and real-time constraints in VLA; action tokenisation versus continuous heads; action chunking; onboard inference on edge accelerators.

Why now. It is the largest piece of writing that depends on nothing, and it will expose whether RQ2 and RQ3 are genuinely open —which is worth discovering before committing.

Recommended order

1. **Send G1–G5 as one short numbered message** to the lab. One message, numbered, easy to reply to. Lead with G1: it is the question whose answer changes the most.
2. **Start T1 and T3 the same day**, without waiting for replies.
3. Fold answers into `00-research-context.md` as they arrive, re-labelling **[UNVERIFIED] → [CONFIRMED]**.
4. Revisit the proposal only if G1 or G2 comes back in a way that changes the lead result.

A draft of the message to the lab is kept locally and is not committed to this repository.